

VANCOUVER HARBOUR, BC, Canada

WMO: 712010

Lat: 49.295N

Lon: 123.122W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -2.6	(c) -0.8	(d) -12.3	(e) 1.3	(f) 0.2	(g) -8.5	(h) 1.8	(i) 1.3	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	26.0	18.1	24.5	17.6	23.2	17.0	19.0	24.5	18.3	23.5	17.6	22.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	11.8	21.7	16.0	11.3	20.8	15.4	10.9	20.1	53.8	24.7	51.6	23.6	49.5	22.7	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -4.5	(f) 29.2	(g) 2.4	(h) 1.7	(i) -6.2	(j) 30.4	(k) -7.6	(l) 31.4	(m) -8.9	(n) 32.3	(o) -10.6	(p) 33.6		
(4) N/A	(5) N/A	(6) N/A	(e) -6.4	(f) 20.2	(g) 2.0	(h) 0.9	(i) -7.9	(j) 20.8	(k) -9.0	(l) 21.3	(m) -10.1	(n) 21.8	(o) -11.5	(p) 22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	5.2	5.7	7.5	10.2	13.7	16.3	18.9	18.8	15.6	11.0	7.4	5.0
	DBStd	5.57	2.91	2.48	2.29	2.24	2.51	2.33	2.08	1.84	2.07	2.08	2.74	2.84
	HDD10.0	633	149	122	83	24	2	0	0	0	0	14	85	155
	HDD18.3	2646	407	355	337	245	143	69	18	17	84	228	329	413
	CDD10.0	1109	1	1	4	29	118	189	275	272	168	44	6	1
	CDD18.3	76	0	0	0	0	1	8	34	30	3	0	0	0
	CDH23.3	278	0	0	0	0	5	39	125	98	11	0	0	0
	CDH26.7	27	0	0	0	0	0	3	17	6	1	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	2242	317	207	215	153	105	86	53	72	107	255	347	324
	PrecMax	2835	529	408	401	314	173	168	127	320	234	435	654	469
	PrecMin	1781	177	11	50	56	26	26	12	10	15	41	157	201
	PrecStd	254	83	82	70	58	35	40	30	66	59	93	122	67
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	12.0	15.2	19.6	23.9	26.7	28.6	27.4	24.7	18.0	14.0	12.0
		MCWB	9.5	9.8	9.7	12.7	15.3	17.2	18.9	18.3	17.3	13.5	12.0	10.1
	2%	DB	10.4	10.6	13.1	17.1	21.7	24.3	26.2	25.7	22.5	16.1	12.7	10.3
		MCWB	8.5	8.2	9.0	11.8	14.8	17.2	18.3	18.2	16.4	12.4	11.0	8.5
	5%	DB	9.6	9.7	11.7	15.4	19.8	22.3	24.5	24.3	20.8	14.8	11.7	9.3
		MCWB	8.0	7.7	8.3	10.6	14.1	16.2	17.6	17.8	15.5	12.1	10.3	7.8
	10%	DB	8.9	8.9	10.6	13.9	18.2	20.7	23.2	22.9	19.3	13.9	10.9	8.6
		MCWB	7.4	7.2	7.9	9.7	13.3	15.3	17.1	17.1	14.9	11.6	9.6	7.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	10.2	11.2	13.7	16.6	18.9	20.6	19.7	18.5	15.0	12.7	10.7
		MCDB	10.6	11.2	14.3	18.1	22.1	24.7	28.2	26.0	23.1	17.2	13.5	11.8
	2%	WB	8.9	9.1	9.9	12.2	15.6	17.8	19.1	18.9	17.2	13.7	11.3	8.9
		MCDB	9.7	10.2	12.4	15.9	20.8	23.1	24.8	24.5	21.2	15.3	12.3	9.8
	5%	WB	8.1	8.2	9.1	11.1	14.7	16.6	18.2	18.3	16.2	12.7	10.5	8.0
		MCDB	9.1	9.4	11.2	14.5	19.2	21.7	23.7	23.5	19.5	14.0	11.4	9.1
	10%	WB	7.4	7.4	8.4	10.2	13.9	15.6	17.6	17.7	15.4	11.9	9.7	7.2
		MCDB	8.4	8.7	10.3	12.7	17.9	20.0	22.7	22.3	18.2	13.5	10.6	8.2
Mean Daily Temperature Range	5% DB	MDBR	3.5	4.4	5.1	6.2	6.9	6.8	7.5	7.4	6.4	4.6	3.8	3.4
		MCDBR	3.9	5.1	6.6	8.7	9.9	9.8	9.9	10.0	9.1	5.9	4.0	3.6
		MCWBR	3.0	3.4	4.1	4.6	4.9	4.4	4.3	4.2	4.3	3.6	3.2	2.7
	5% WB	MCDBR	3.6	4.0	6.2	7.3	8.8	9.1	9.5	9.3	7.6	4.8	3.7	3.3
		MCWBR	3.0	3.0	4.0	4.3	4.6	4.4	4.2	4.2	3.9	3.1	3.0	2.7
Clear-Sky Solar Irradiance	taub	0.319	0.313	0.331	0.366	0.391	0.367	0.364	0.368	0.350	0.339	0.323	0.307	
	taud	2.405	2.472	2.443	2.367	2.340	2.434	2.465	2.473	2.523	2.534	2.487	2.429	
	Ebn at Noon	730	842	883	881	869	892	890	871	853	801	726	694	
	Edh at Noon	66	77	93	111	120	110	105	100	86	72	61	56	
All-Sky Solar Radiation	RadAvg	0.85	1.70	2.54	3.97	5.21	5.58	6.25	5.31	3.67	2.01	0.99	0.68	
	RadStd	0.14	0.26	0.36	0.44	0.46	0.68	0.57	0.43	0.33	0.29	0.19	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+18
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page